

# Metaheuristic Optimization for Vehicle Routing and V2G Scheduling in Electrical Vehicle Fleet Network

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**Abstract**—The paper investigates the Vehicle Routing and Charging/Discharging Scheduling Problem for an electric vehicle fleet with Vehicle-to-Grid (V2G) technology for a given network. Moreover, it compares different algorithms to maximize the revenue generated by V2G through the charging and discharging of electric vehicles, while ensuring they reach their destination by a given deadline. In order to reduce the high complexity of the problem, four different metaheuristic algorithms are introduced: Simulated Annealing (SA), Genetic Algorithm (GA), Discrete Grey Wolf Optimizer (DGWO), and Shuffled Frog Leaping Algorithm (SFLA). The performance of the implemented metaheuristic algorithms is also examined through multiple experiments with different tuning parameter. The performance of each algorithm has been found to vary significantly. The research also provides insights in effective configuration for the Vehicle Routing and Charging/Discharging Scheduling Problem for Vehicle-to-Grid with a focus on profit maximization.

**Index Terms**—Vehicle routing, electric vehicles, vehicle-to-grid (V2G), charging and discharging scheduling, metaheuristic optimization, genetic algorithm (GA), simulated annealing (SA), discrete grey wolf optimizer (DGWO), shuffled frog leaping algorithm (SFLA), fleet optimization.

## I. INTRODUCTION AND RELATED WORK

THE transportation sector makes a considerable contribution to gas emissions and pollution caused by the high level of petroleum use. In order to curb pollution and emissions, the use of electric vehicles has been cited as a possible replacement for conventional vehicles driven by internal combustion engines. The rapid growth of EV fleets will also imply a rise in electricity demand, thus exerting further pressure on the existing infrastructure to meet the increasing demand for electricity.

Recent breakthroughs in vehicle-to-grid technology also provide a possible answer to these issues. Vehicle-to-grid is a system of energy transfer between EVs and the grid. The EVs can use the grid for their operation and also feed the grid with energy as required. The EVs thus change from being energy consumers to being energy resources for the grid, thus providing sustainability to the environment.

Despite the positive attributes of V2G, the majority of studies on the issue assume that EVs are stationary and are always available for interaction with the grid at any given point in time and location. However, EVs are often mobile, especially those deployed for fleet operations such as delivery

services, ride-sharing services, and public transport services. In order to address this problem, recent research has combined vehicle routing with the energy management of EVs with V2G technology. The research has been categorized into different classes: exact/hybrid, heuristic, and learning-based approaches, with the aim of maximizing profit.

### Literature-based Approaches:

The problem has been formulated as an Electric Vehicle Routing Problem with Time Windows – Time Expanded formulation (EVRPTW-TP) and has been addressed using exact and hybrid optimization methods. This formulation encloses the following: the routing problem, charging and discharging operations, dynamic battery status for the electric vehicles, and time windows, which act as deadlines for the deployed vehicles. In the work proposed by Lin et al. [1], Lagrangian Relaxation has been combined with the Variable Neighborhood Search/Tabu Search (VNS/TS) approach to formulate dynamic pricing and station capacities. This approach has achieved promising results but has shown high computational complexity. In another study, Liu et al. [2] have proposed a two-stage approach combining the K-Shortest Path and Integer Programming to solve the joint problem, but the approach has shown limited scalability and coordination. To reduce the computational complexity and achieve faster results, learning-based and heuristic approaches have also been incorporated into the problem formulation. For example, reinforcement learning-based approaches, such as QuikRouteFinder [3], have achieved faster results but may also show sub-optimality and limited interpretability. Evolutionary optimization has also been considered in the context of the problem formulation. For example, Du et al. [4] have proposed an orienteering-based vehicle-to-grid approach to achieve increased profit, but only in single-vehicle scenarios. Very recently, cooperative approaches have also been considered in the context of the problem formulation. For example, Liu et al. [5] have proposed cooperative game theory-based distributed coordination to achieve enhanced performance.

### Research gap:

Many approaches were proposed to solve this problem, with the formulation enclosing the routing problem, charging and discharging operations, dynamic battery levels for the EVs, and deadlines for the deployed vehicles. However, the recent developments of trajectory-based and population-based metaheuristics for integrated and cooperative multi-EV routing problems with V2Gs have not yet been thoroughly and extensively explored. Most of the existing research on metaheuristics is only applicable to single vehicle scenarios or assumes that energy trading is not integrated with routing

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decisions, which is again not a scalable and coordinated solution approach for a very combinatorial yet multi-objective problem of cooperative V2Gs.

Accordingly, a comparative assessment of four metaheuristics, including one trajectory-based technique (Simulated Annealing) and three population-based techniques (Genetic Algorithm, Discrete Grey Wolf Optimizer, and Shuffled Frog Leaping Algorithm), is presented in this paper.

## II. METHODOLOGY

### A. Problem Formulation

This paper aims to optimize the routing and energy trade scheduling for a fleet of electric vehicles (EVs) operating within a vehicle-to-grid (V2G) transportation network to maximize total trading profit and on-time arrivals.

The transportation and energy network is modeled as an undirected, weighted, fully connected graph  $G = (\mathcal{N}, \mathcal{E})$ , where any node can be reached from any other node in the graph. Nodes can either represent a start point, a destination, or a power station. Every power station in the subset of power station nodes  $\mathcal{S}$  is characterized by charging/discharging rates and limited service capacity  $K_u$ . Nodes are connected through edges where every edge  $(u, v)$  is assigned a specific travel time  $\tau_{uv}$  and an associated energy consumption value  $e_{uv}$ . A distance matrix is created to represent the graph connectivity and spatial relationships between all nodes. This matrix represents the temporal and energetic costs of moving from one node to another.

Each EV must obtain a path from start to destination while passing through intermediate power station nodes. An EV  $a \in \mathcal{A}$  is defined by its start node  $o_a$ , destination  $d_a$ , deadline  $\Delta_a$ , and battery capacity  $B_a^{\max}$ . Each EV's objective is to reach its destination before the assigned deadline. The EV fleet is heterogeneous with respect to battery capacities, deadlines, and destinations. At each power station, an EV can charge, discharge, or pass by the station depending on its current battery level and the fixed energy prices at each station.

Some assumptions were followed for accurate modeling. The EV fleet operate in discrete time and start with the maximum battery capacity, which is assumed to be greater than the shortest distance between any two nodes. Furthermore, every time unit costs 1 unit of energy, and all stations have the same rate of charging and discharging (1 unit of time charges/discharges 1 unit of energy). In addition, the charging cost is greater than the discharging cost. Finally, a power station has no limit on the energy it can sell or buy, and an unlimited number of EVs can pass through any edge.

The decision variables of the defined problem include the path of every EV,  $\mathcal{H}_a^{\text{path}}$ , which starts from the specified starting node  $o_a$ , passes through intermediate nodes, and ends at the specified destination node  $d_a$ :

$$\mathcal{H}_a^{\text{path}} = [o_a, u_{a,1}, \dots, u_{a,n}, d_a], \quad (1)$$

where  $u_{a,n}$  is the  $n$ -th node visited by agent  $a$ . The decision variables also include a battery array,  $\mathcal{H}_a^{\text{battery}}$ , which stores the battery capacity of the EV at every node:

$$\mathcal{H}_a^{\text{battery}} = [B_a^{\max}, b_{a,1}, b_{a,2}, \dots, b_{a,n}], \quad (2)$$

where  $b_{a,n}$  denotes the battery level upon leaving node  $u_{a,n}$ . It is vital to note that the length of  $\mathcal{H}_a^{\text{path}}$  and  $\mathcal{H}_a^{\text{battery}}$  is a fixed constant to be determined before running any optimization algorithm. The battery level implicitly decides the action to be performed at the node  $u_{a,n}$ .

This optimization problem addresses two main objectives: maximizing the total number of EVs that reach their specified destination before their specified deadline, presented by equation [3], and maximizing the total revenue collected by all EVs from trading with the grid, as shown in equation [4]. The primary objective function that maximizes successful arrivals is:

$$\max \sum_{a \in \mathcal{A}} z_a \quad (3)$$

where  $z_a = 1$  if EV  $a$  reaches its destination before its deadline. The secondary objective function, which maximizes economic profit, is:

$$\max \sum_{a \in \mathcal{A}} z_a \sum_{u \in \mathcal{S}} (p_u^{\text{sell}} E_{a,u}^{\text{dis}} - p_u^{\text{buy}} E_{a,u}^{\text{ch}}), \quad (4)$$

where  $p_u^{\text{sell}}$  and  $p_u^{\text{buy}}$  represent the energy selling and buying prices at station  $u$ , respectively, and  $E_{a,u}^{\text{ch}}$  and  $E_{a,u}^{\text{dis}}$  represent the amount charged and discharged by EV  $a$  at station  $u$ . A vehicle that runs out of battery or fails to reach its destination before the deadline is excluded from the profit objective function. The combined multi-objective formulation is

$$\max w_1 \sum_a z_a + w_2 \sum_a z_a \sum_{u \in \mathcal{S}} (p_u^{\text{sell}} E_{a,u}^{\text{dis}} - p_u^{\text{buy}} E_{a,u}^{\text{ch}}), \quad (5)$$

where  $w_1, w_2$  define the weight of importance of timely arrival, and profit gained. Usually, the following relation is followed to signify the influence of the primary objective function:

$$w_1 > w_2. \quad (6)$$

The problem discussed in this study is restricted by three different constraints:

- 1) **Energy Dynamic Constraints:** Charging increases the battery capacity, discharging and traveling along edges decrease the battery capacity, given this equation:

$$0 \leq B_a(t) \leq B_a^{\max}, \quad \forall a \in \mathcal{A}, t \in \mathcal{T} \quad (7)$$

where  $\mathcal{T}$  is the discrete time horizon,  $B_a(t)$  is the battery level of agent  $a$  at time  $t$  and  $B_a^{\max}$  is the battery capacity of agent  $a$ .

- 2) **Action Exclusivity and Mutual Exclusion:** Each EV can perform only one action at a station: charging, discharging, or passing through—at any given time.
- 3) **Station Capacity Constraint:** Each charging station has a finite, predefined number of available slots where every slot can only accommodate one EV at a time, represented by  $K_u$ . The total number of EVs simultaneously charging or discharging cannot exceed this capacity:

$$\sum_{a \in \mathcal{A}} 1\{\text{EV } a \text{ is active at station } u \text{ at time } t\} \leq K_u \quad (8)$$

### B. Experimental Setting

Two different maps were used to set the experimental environment where the optimization solvers are tested.

The first map is a simple, small map that is mainly used to validate and assess the performance of the algorithms, analyze basic decision-making behavior, and observe agents' interactions in a controlled setting. It consists of five nodes (one start node, one destination node, and three power stations).

The second map is a bigger and more complex map that consists of ten nodes (two start nodes, two destination nodes, and six power stations). This map presents a richer connectivity structure between different nodes, giving rise to more routing alternatives, resource contention at charging stations, and scheduling challenges. This map's complexity evaluates the robustness and scalability of the tested algorithms under more realistic operating conditions.

### C. Metaheuristic Solvers

To initialize all of the upcoming algorithms, initial random solutions for all agents were created by a general procedure. For every agent, a random variable defines the number of power stations between a randomly chosen start node and destination node. Simultaneously, a sequence of battery levels is generated to create corresponding actions while strictly maintaining feasibility and confinement to the energy dynamic constraints. To ensure a consistent data structure across all agents for subsequent operations, both paths and battery level sequences are padded to a fixed solution size. A solution illustration is shown in Figure 1.

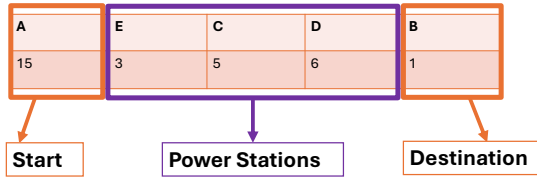


Fig. 1: An example of a solution output showing both the nodes chosen and the corresponding energy level.

1) *Simulated Annealing (SA)*: SA is a trajectory-based metaheuristic algorithm first used in 1983 [6] to solve combinatorial optimization problems by modeling the metallurgical process of physical annealing. The algorithm mimics physical annealing by utilizing a temperature parameter that resembles the tendency to probabilistically accept worse solutions to find better ones, not to be trapped in local maxima. This makes it effective in solving complex problems with large search spaces. It begins with a high temperature  $T_0$  to explore the search space, and it employs a cooling mechanism to decrease the temperature, thereby giving more prominence to exploitation. Neighborhood operators are to maintain solution feasibility by randomly replacing a path node with an alternative power station and adjusting adjacent battery levels to maintain feasibility.

2) *Genetic Algorithm (GA)*: GA is an effective population-based metaheuristic optimization algorithm that utilizes biologically inspired operators to produce successive generations

of candidate solutions emulating the process of evolution. The process is controlled using three different operators: the elitism operator, which passes over the best individuals to the next generation; the crossover operator, which merges segments from the path and battery level array solutions of two high-performing parents to produce potentially better offspring; and the mutation operator, which refines the worst solutions by randomly replacing path nodes and adjusting battery levels if necessary to ensure feasibility. GA has proven to be effective in solving similar task allocation problems [7] and path planning problems [8], [9].

3) *Discrete Grey Wolf Optimizer (DGWO)*: Grey wolf optimization (GWO) is a population-based optimizer that models the wolves' pack natural hierarchy and collective hunting nature, where the best three individuals: Alpha ( $\alpha$ ), Beta ( $\beta$ ), and Delta ( $\delta$ ) wolves are responsible for guiding the rest of the pack, which are known as Omega ( $\omega$ ) wolves. Since GWO is commonly used for continuous problems, a discrete version adopted from the methodology of Panwar and Deep (2021) [10] for the Traveling Salesman Problem was applied with some adjustments to align with the defined problem. A "Move-TowardsLeader" operator was responsible for probabilistically updating the paths of the  $\omega$  wolves by merging segments from the  $\alpha$ ,  $\beta$ , and  $\delta$  path and battery level solution arrays while maintaining the feasibility and compliance of the new merged solutions to energy dynamic constraints. A mutation procedure was probabilistically applied to some  $\omega$  wolves to prevent premature convergence.

4) *Shuffled Frog Leaping Algorithm (SFLA)*: SFLA is also a population-based metaheuristic optimizer that simulates the natural behavior of frogs while searching for food. With reference to the publication titled "Shuffled Frog-Leaping Algorithm: A Memetic Metaheuristic for Discrete Optimization" [11], a modified version of the algorithm was designed to apply to our problem. The population is ranked in descending order and divided into a defined number of memeplexes  $m$ , each having  $n$  frogs. Within each memeplex, smaller sub-memeplexes of variable size  $q$  were formed based on individuals' fitness. The solution of the worst fitness in each sub-memeplex goes in an iterative enhancement attempt by first trying to adopt path and battery level segments from the local best frog in the same memeplex. If this attempt failed to enhance the fitness value, the solution is then guided toward the global best frog ( $\alpha$ ). If both attempts failed, a random re-initialization procedure is applied to maintain population diversity. This process is repeated for a fixed number of times, then all memeplexes are shuffled to allow for global information change.

## III. RESULTS AND DISCUSSION

Table I presents the parameter configurations tested for SA, GA, DGWO, and SFLA. The experiments examined how population size, operator distribution, cooling schedules, and iteration limits affect convergence behavior, solution stability, and performance as problem complexity increases.

TABLE I: Evaluated Parameter Configurations

| Algorithm | Parameters Tested                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| SA        | $T_0 = \{1000, 500\}$ , $\alpha = \{0.95, 0.20\}$ ,<br>Iterations = $\{1000, 500\}$                                           |
| GA        | Population = $\{100, 50\}$ , Generations = 100,<br>Rates = (60% crossover/mutation/elitism),<br>Equal distribution (33% each) |
| DGWO      | Population = $\{100, 50\}$ , Iterations = 100,<br>Mutation = 10%                                                              |
| SFLA      | Population = $\{100, 50\}$ ,<br>Memeplex parameters $(m, n)$ (low/high)                                                       |

#### A. Parameter Impact on Algorithm Performance

1) *Simulated Annealing (SA)*: SA is strongly affected by the temperature settings. For the experiment with only two EVs, high initial temperature ( $T_0 = 1000$ ) with slow cooling ( $\alpha = 0.20$ ) results in stable results (mean fitness 278.30, standard deviation 5.26). This experiment's settings enabled broader exploration of the search space. For the other experiment with 5 EVs instead of two, the optimal parameter configuration was the opposite: low temperature with fast cooling ( $\alpha = 0.95$ ) performed better (mean fitness 584.40, standard deviation 7.58). This indicates that exploration becomes a less important factor in complex search spaces.

Despite this, SA performance degraded on larger maps. The performance decreased by 58-86%. SA is not a population-based algorithm. For this reason, the single solution approach causes severe scalability limitations on larger maps with a high number of EVs and frequent convergence to infeasible solutions. Despite achieving the fastest computation times (0.31 seconds), the trajectory-based search struggled with multi-vehicle interactions and combinatorial constraints, making it unsuitable for complex V2G routing scenarios.

2) *Genetic Algorithm (GA)*: GA consistently achieved better performance with larger population sizes. Increasing the population from 50 to 100 reduced variance and improved both optimal and average fitness values. For 2 EVs on small maps, equal operator rates (33% each) achieved the best results (mean fitness 291.55, standard deviation 5.04). For 5 EVs, a high crossover rate (60%) shows a superior performance (mean fitness 680.97, standard deviation 9.94).

These results show that the crossover rate is critical for generating high-accuracy solutions in multi-EV routing problems. As the complexity of the problem grew, the GA continued to converge reliably. Low standard deviation (15.97) and 100% success rate assure consistent performance across all trials. The convergence graph shows that the GA found a fairly good solution within the first 20 iterations while continuing to enhance it throughout the search process.

3) *Discrete Grey Wolf Optimizer (DGWO)*: DGWO showed improved performance with population size 100 compared to 50 (mean fitness 659.83, standard deviation 17.70 versus mean fitness 649.27, standard deviation 28.53 for Map 1 with 5 EVs). DGWO's hierarchical leadership structure facilitates rapid convergence in early iterations, resulting in better initial solutions compared to the GA. Nevertheless, the decrease in population diversity in the following iterations makes it susceptible to falling into a local optimum when dealing with

large maps with a large number of EVs. While the DGWO showed strong success rates (around 95%) and resulted in solutions of comparable quality, its initial convergence speed benefits became less effective as map size and complexity increased.

4) *Shuffled Frog-Leaping Algorithm (SFLA)*: For small maps, high population (population size = 100) achieved the best performance with a mean fitness of around 662 and a maximum profit of around 162. The configuration  $m = 20, n = 5$  produced the most stable results (standard deviation approximately 13.2). On large maps, performance decreased due to increased complexity, with mean fitness ranging from 550 to 578 and profit between 50 and 65.

SFLA shows steady improvement over the iterations. With its memplex structure, it shows robust exploration capabilities. Larger population sizes consistently outperformed smaller ones, confirming the importance of population size for maintaining well performance solution. While SFLA's solution quality was competitive, it remained slightly lower than GA and DGWO, and required longer computation times (1.90 seconds).

#### B. Comparative Performance Analysis

Table II compares algorithm performance on Map 1 with 5 EVs across 20 independent runs.

TABLE II: Algorithm Performance Comparison (Map 1, 5 EVs, 20 Runs)

| Metric           | SA          | GA            | DGWO   | SFLA   |
|------------------|-------------|---------------|--------|--------|
| Optimal Fitness  | 573.50      | <b>709.70</b> | 693.40 | 688.00 |
| Best Profit      | 73.50       | <b>209.70</b> | 193.40 | 188.00 |
| Mean Fitness     | 522.96      | <b>689.83</b> | 636.78 | 654.12 |
| Std. Deviation   | 40.71       | <b>15.97</b>  | 22.59  | 23.16  |
| Success Rate (%) | 35.7        | <b>100.0</b>  | 95.2   | 95.0   |
| Comp. Time (s)   | <b>0.31</b> | 1.71          | 1.40   | 1.90   |

GA achieved superior performance across all evaluated factors. It obtained the highest optimal fitness (709.70), best profit (209.70), and mean fitness (689.83), exceeding DGWO by 16.3 units and SA by 136.2 units in optimal fitness. GA was the only algorithm to achieve 100% success rate, demonstrating complete reliability across all runs. Its low standard deviation (15.97) indicates high consistency and low variability.

DGWO and SFLA delivered competitive results with success rates above 95%, demonstrating reliable performance for cooperative multi-EV V2G routing. However, both demonstrate higher variability than GA. DGWO's computational efficiency (1.40 seconds) positioned it between GA and SFLA, offering a balanced trade-off between solution quality and execution time.

SA required the lowest computation time (0.31 seconds), but produced significantly weaker solutions with poor reliability. The major quality reduction and high failure rate (64.3%) make SA unsuitable for complex multi-EV scheduling problems, despite its speed strength.

### C. Convergence Behavior Analysis

Figure 2 shows the convergence patterns for all four algorithms over 100 iterations across 20 runs, and the following can be observed:

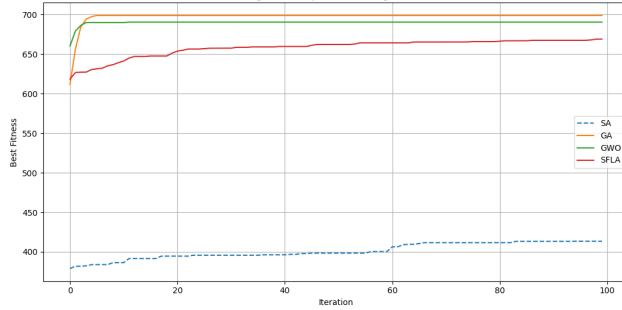


Fig. 2: Convergence curves for SA, GA, DGWO, and SFLA showing best fitness values over 100 iterations across 20 runs.

1) *Solution Quality*: GA converged to the highest final fitness value (approximately 690), which is consistent with the observations from Table II. DGWO and SFLA converged to competitive but lower values (approximately 690 and 670, respectively). However, SA converged to a significantly lower fitness value (under 420), confirming its poor solution quality for the defined optimization problem.

2) *Convergence Speed*: GA and DGWO show rapid initial convergence, reaching close-to-optimal fitness within the first 20 iterations. DGWO showed slightly faster initial improvement, and its hierarchical leadership mechanism supports this behavior. SFLA demonstrates steady, gradual improvement from iteration 20 to 60, showing continuous exploration and balanced searching. SA shows slow, inconsistent progress over the 100 iterations, with repeated periods of no improvement at all.

3) *Search Dynamics and Exploration-Exploitation Balance*: The rapid convergence and plateau of GA and DGWO curves indicate strong exploitation of high-quality solutions found early in the search phase. GA found the best solution by iteration 20 and remained stable, which highlights a superior balance between exploration and exploitation, supported by effective genetic operators tuning.

SFLA's improvement and overall convergence took longer than other algorithms, suggesting a more consistent exploration process through its memplex-based local search mechanism. The gradual improvement pattern indicates the ability to escape being stuck in local optima through the global information exchange process between memplexes.

SA's low-fitness curve indicates poor exploration and exploitation. It can be concluded that the trajectory-based approach failed to explore the large, complex multi-agent solution space, causing the algorithm to get stuck in sub-optimal solutions.

### D. Performance Trade-offs and Implications

The comparative analysis reveals important trade-offs between solution quality, computational efficiency, and robustness:

**Solution Quality vs. Computational Time**: While SA provides the fastest execution time (0.31 seconds) it exhibits lower solution quality. Conversely, GA achieves highest solution quality at a modest computational time (1.71 seconds). This trade-off emphasizes that the extra computational requirements of population-based approaches is justified for realistic V2G problems, where solution quality directly affects financial gain and grid stability.

**Reliability vs. Speed**: The fact that GA's success rate is 100% whereas SA's success rate is 35.7% shows how important dependability is for practical uses. Re-optimization is necessary for failed solutions, which could offset any time savings from faster algorithms.

**Scalability Considerations**: The performance difference between trajectory-based algorithms (SA) and population-based algorithms (GA, DGWO, SFLA) grew dramatically as the problem complexity increased (bigger maps, more agents). This shows that in order to scale to realistic fleet sizes and network topologies, population-based techniques are necessary and highly effective. Their ability to explore large, multi-modal solution spaces and maintain feasibility under operational constraints such as battery capacities, station service limits, and time windows is critical for practical deployment in the real-world.

### E. Algorithm Selection Guidelines

Based on the experimental findings, the following recommendations apply for practical V2G routing deployment:

- **Genetic Algorithm (GA)**: Recommended for large-scale, complex multi-EV routing problems where solution quality and reliability are critical. Use a population size of 100 and a 60% crossover rate for optimal performance. Best suited for mission-critical applications requiring maximum economic profit and grid stability.
- **Discrete Grey Wolf Optimizer (DGWO)**: Suitable for moderate-complexity scenarios requiring a balance between solution quality and computational efficiency. Use a population size of 100 to maintain stability and mitigate premature convergence to local optima. Appropriate when faster convergence is valued.
- **Shuffled Frog-Leaping Algorithm (SFLA)**: Appropriate for applications requiring sustained refinement and robust exploration. Configuration with  $m > n$  and a population size of 100 provides the best consistency. Suitable when extended search time is acceptable.
- **Simulated Annealing (SA)**: Only suitable for small-scale problems ( $\leq 2$  EVs) with strict time constraints and when solution quality requirements are minimal. Not recommended for practical V2G fleet management due to poor reliability and scalability limitations.

### F. Analysis of the Best Solution

The following figures summarize the performance of the best solution obtained by GA. Figure 3 depicts the resulting routes taken by all EVs towards the destination. Figure 4 presents the EVs' actions over time, reflecting the energy-related decisions made throughout the paths. Finally, Figure

5 provides a breakdown of the achieved profit, showing how routing strategies and charging/discharging decisions affect the overall system performance.

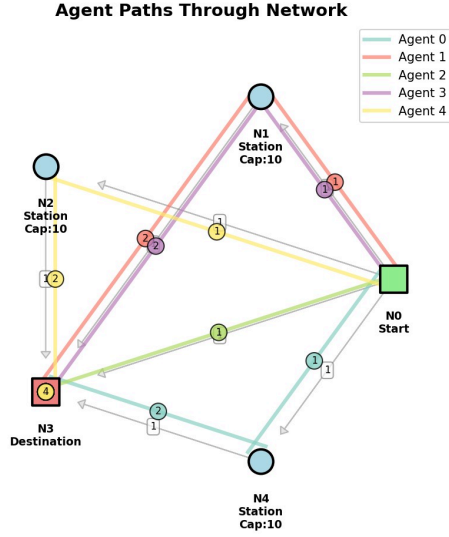


Fig. 3: Optimized trajectories of all EVs acquired by the best GA solution.

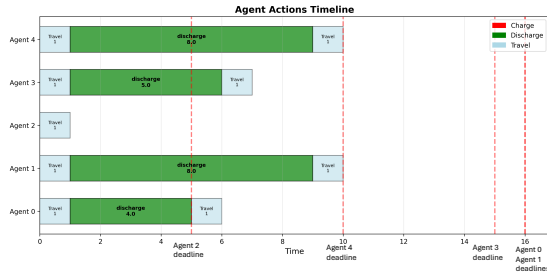


Fig. 4: EV actions over time for the best GA solution, demonstrating charging, discharging, and movement decisions.

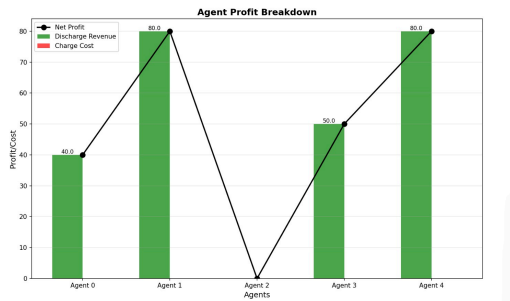


Fig. 5: Profit analysis achieved by the best GA solution, focusing on the contribution of each EV in the fleet.

#### IV. CONCLUSION AND FUTURE WORK

This study investigated joint routing and vehicle-to-grid scheduling optimization for an electric vehicle fleet using

metaheuristic algorithms. Four algorithms—Simulated Annealing (SA), Genetic Algorithm (GA), Discrete Grey Wolf Optimizer (DGWO), and Shuffled Frog-Leaping Algorithm (SFLA)—were evaluated on EV fleet coordination problems of varying complexity.

The experimental results confirm that population-based metaheuristics, particularly GA, provide reliable and scalable solutions for integrated routing and V2G scheduling in multi-EV systems. GA consistently achieved the highest solution quality, optimal fitness values, and profit with perfect reliability (100% success rate) and minimal variability. DGWO and SFLA demonstrated competitive performance with high success rates ( $\approx 95\%$ ) and stable convergence behavior. On the other hand, SA's trajectory-based approach proved unsuitable for complex multi-EV scenarios despite faster execution times.

While this study demonstrates the effectiveness of metaheuristic optimization for joint EV routing and V2G scheduling, several directions remain for further investigation to enhance realism, scalability, and practical deployment.

- Extending the framework to real-world maps and traffic scenarios.
- Enabling adaptive and scalable decision-making under dynamic conditions, including time-varying traffic patterns and fluctuating grid prices.
- Integration of dynamic traffic conditions and real-time energy pricing.

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